



भारतीय प्रौद्योगिकी
संस्थान जम्मू
INDIAN INSTITUTE OF
TECHNOLOGY JAMMU

B. Tech. CS/EE/MT/MA (1st SEM)/ Mid-Semester Examination 2025-26

Code & Course: CE-1-02(PO) & CAD

Date: 13/09/2025 (Saturday)

Duration: 2 Hours

Time: 10:00 am – 12:00 pm

Total marks: 30

SET A

Note

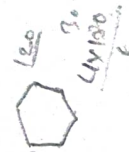
- Mobile phones are not allowed into the lab during the examination. If found, disciplinary action will be taken as per the ordinance.
- Talking, exchange of verbal/written communication of any type, will be treated as malpractice and will be dealt with accordingly.
- The full access link of the "Onshape" document shall be shared **only with your instructor before 10.10 AM**, and you shall create the sketch in that document during the examination. No other document will be accepted for submission and will not be evaluated.
- **Email IDs** of instructors for sharing the Onshape document:
 - Computer Science and Engineering: angshuman.kapil@iitjammu.ac.in
 - Electrical Engineering: chembolu.vinay@iitjammu.ac.in
 - Materials Engineering: akash.awale@iitjammu.ac.in
 - Mathematics and Computing: pratik.kumar@iitjammu.ac.in

Before you start, read the following instructions carefully:

- All dimensions are in **mm**.
- Assume suitable dimensions for missing ones, if any.
- After the exam time is over, **DO NOT** edit your Onshape sketch, and **DO NOT** make any changes.

Questions:

Q1. A line AB measuring 70 mm has its end A 20 mm in front of VP and 20 mm above HP and the other end B 60 mm in front of VP and 50 mm above HP. Draw the projections of the line and find the inclinations of the line with the both the reference planes of projection.	[5]
Q2. Draw the projections of a regular hexagon with <u>35 mm</u> sides, ensuring that one corner is positioned on the VP. The corner opposite to this corner is <u>40 mm</u> in front of the VP. The front view of the diagonal through these corners creates a 45° inclination with H.P.	[7.5]

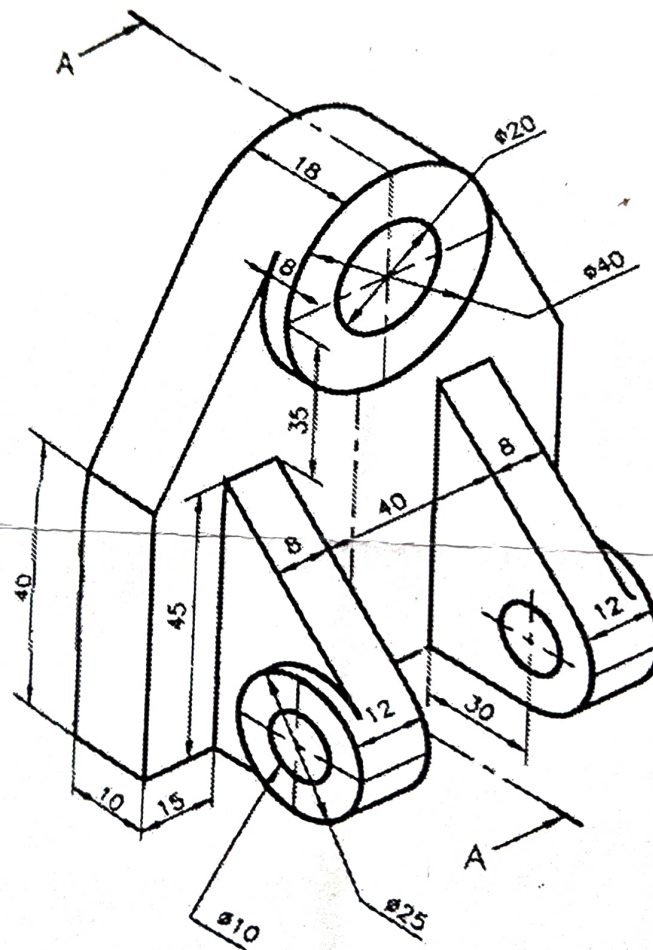




Q3. Using the First Angle Projection Method, Draw the Front View (FV), Top View (TV) and Left Hand Side View (LHSV) of the object given in Figure 1.

[7.5]

Figure 1:



TV

FV

S.V



$$15 + 16 + 10$$

$$= 41$$

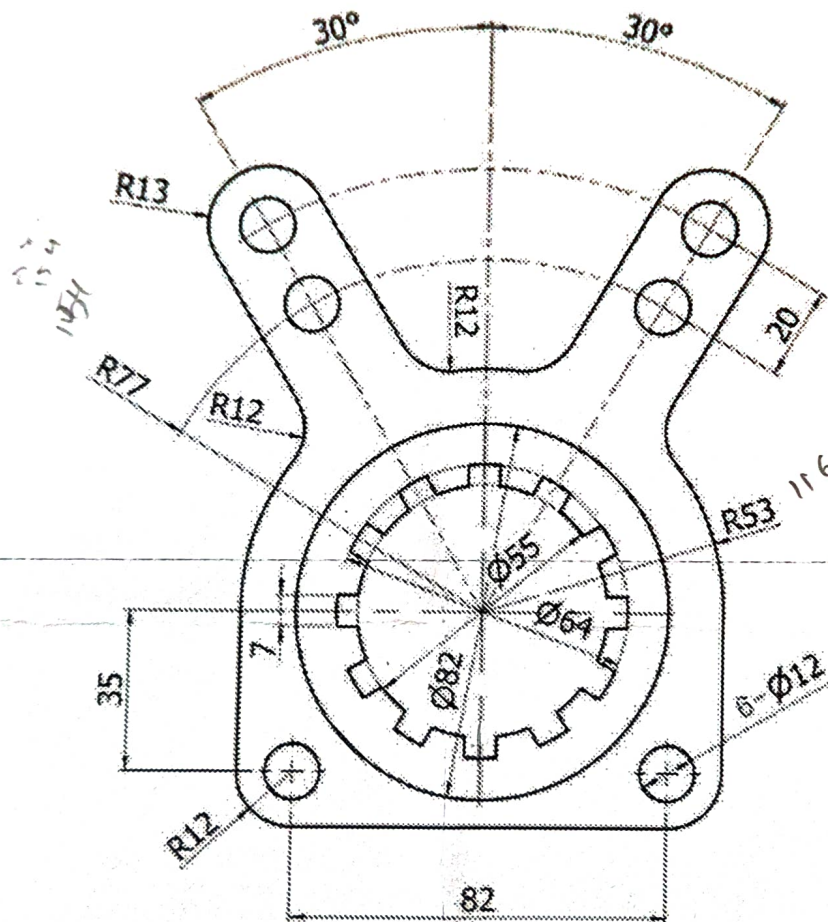
40



Q4. Using Onshape, Create the fully constrained sketch of the Figure 2 provided below.

[10]

Figure 2



End of the Paper



भारतीय प्रौद्योगिकी
संस्थान जम्मू
INDIAN INSTITUTE OF
TECHNOLOGY JAMMU

B.Tech. CSE/MNC/EE/MT (I SEM)/ End Semester Examination 2025-26

Code & Course: CE-1-02(PO) & CAD

Duration: 3 Hours
Time: 10:00 am – 1:00 pm

Date: 15/11/2025 (Saturday)

Note

- Mobile phones are not allowed into the lab during the examination. If found, disciplinary action as per the ordinance will be taken.
- Talking, exchange of verbal/written communication of any type, will be treated as malpractice and will be dealt with accordingly.
- The full access link of the "Onshape" document shall be shared **only with your instructor before 10.10 AM**, and you shall create parts and an assembly in that document during the examination. No other document will be accepted for submission and will not be evaluated.
- Email IDs of instructors for sharing of Onshape document:
 - MT: akash.awale@iitjammu.ac.in
 - EE: chembolu.vinay@iitjammu.ac.in
 - CSE: angshuman.kapil@iitjammu.ac.in
 - MNC: pratik.kumar@iitjammu.ac.in

Before you start, read the following instructions carefully:

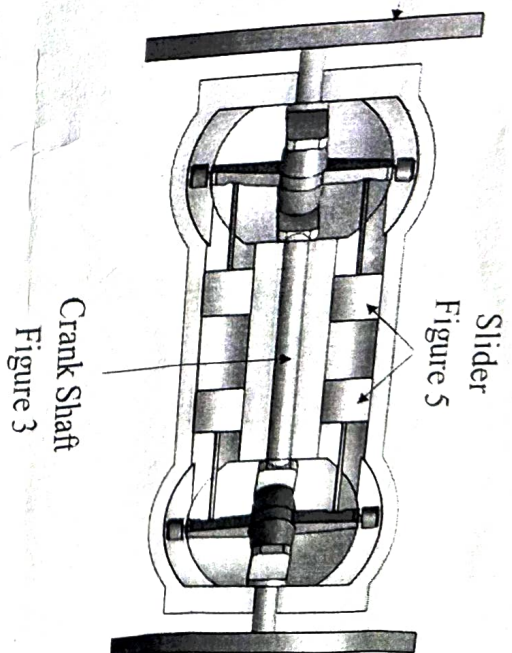
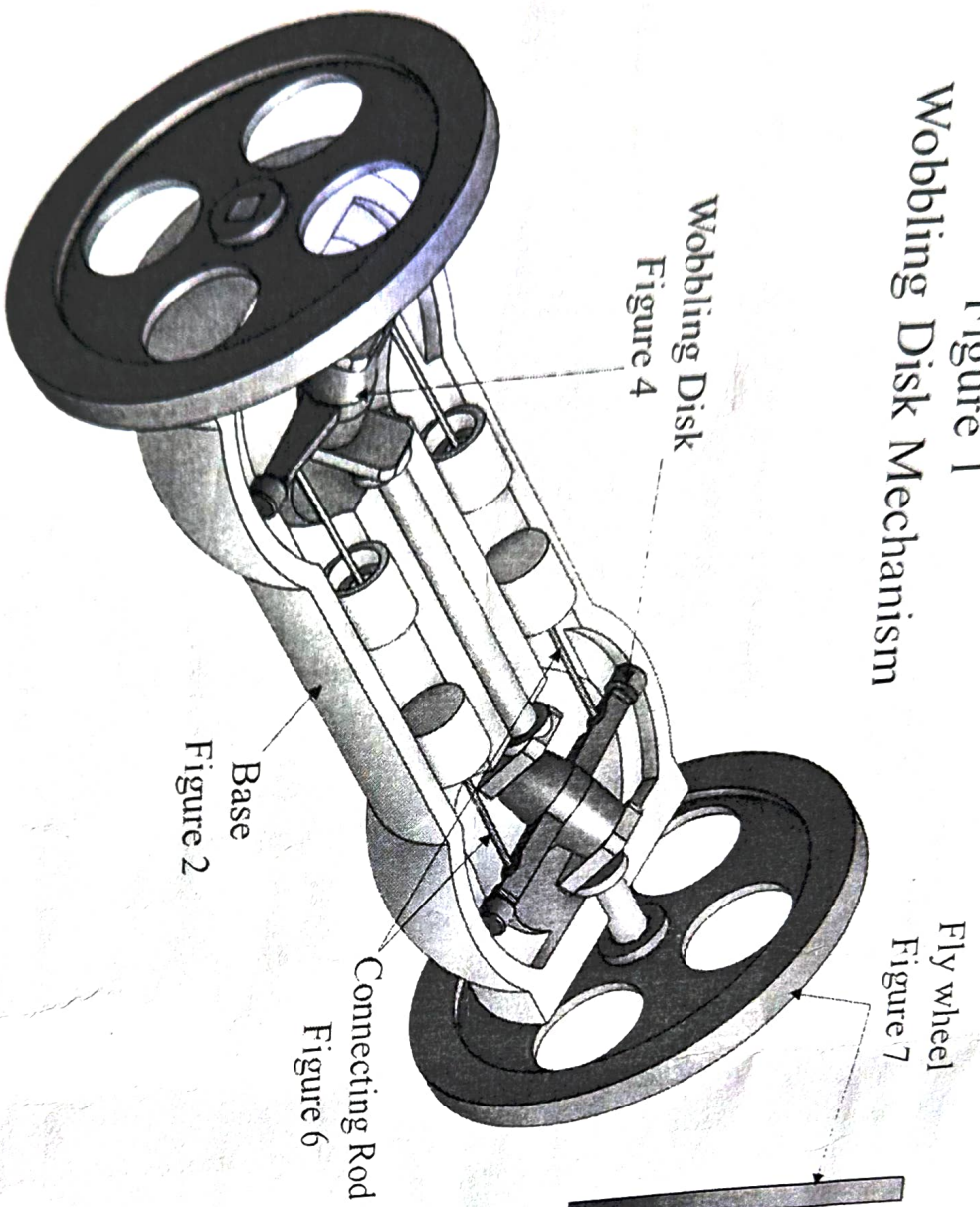
- The assessment of the submitted Onshape document for the end semester examination will be based on two major components: **part modeling** and **assembly** (including **animation**), each carrying specific weightage.
- All dimensions are in **mm**, and all drawings are in the **First Angle** system of projection.
- Assume suitable dimensions for missing ones, if any.
- After the exam time is over, **DO NOT edit your model**, and **DO NOT make any changes**.
- Additional figures and videos are given to help realize the assembly. The link for the same is <https://tinyurl.com/3k6dn382>

Question:

Create the assembly of a **Wobbling Disk Mechanism** (Figure 1). The drawings of individual parts (Figure 2 to Figure 7) are provided below. Also, demonstrate the functionality of the assembly as per the animation provided.

*****BEST OF LUCK*****

Figure 1
Wobbling Disk Mechanism



19
35

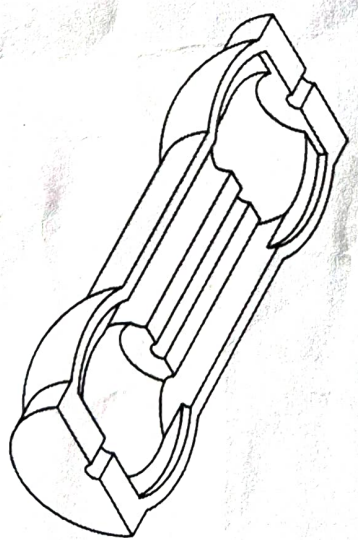
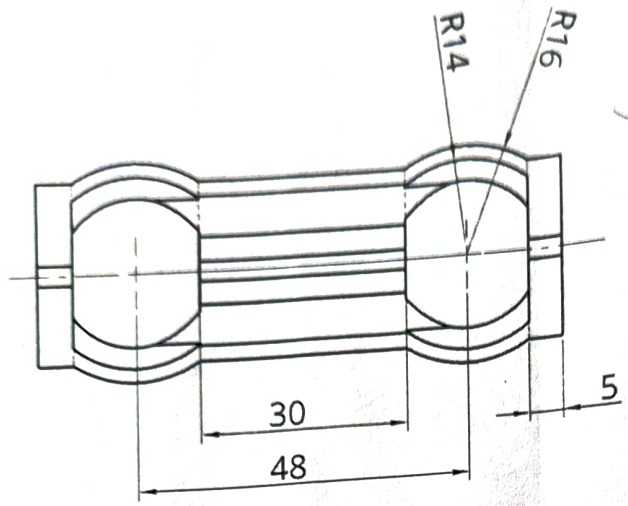
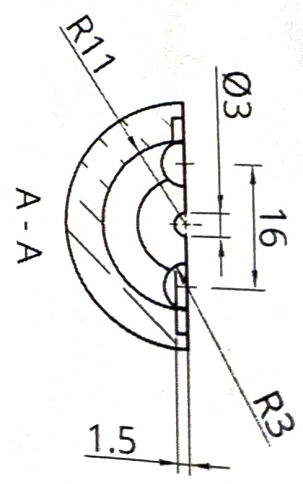
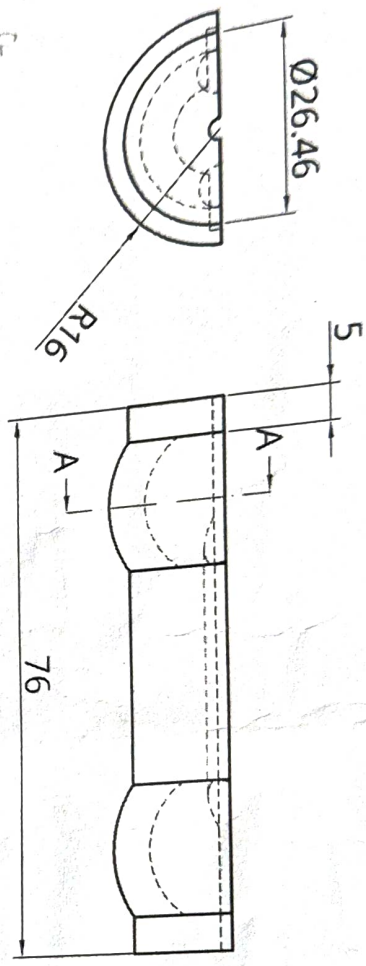
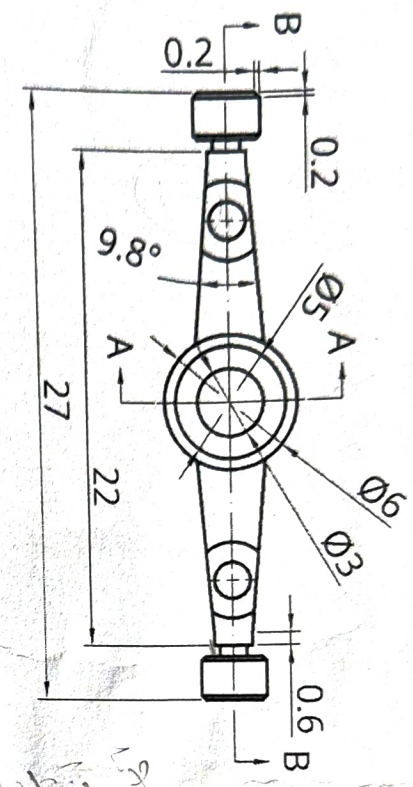


Figure 2
Base
Qty 1



1.94
5.94
3.48

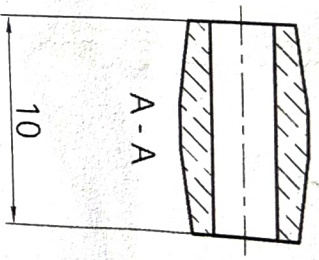
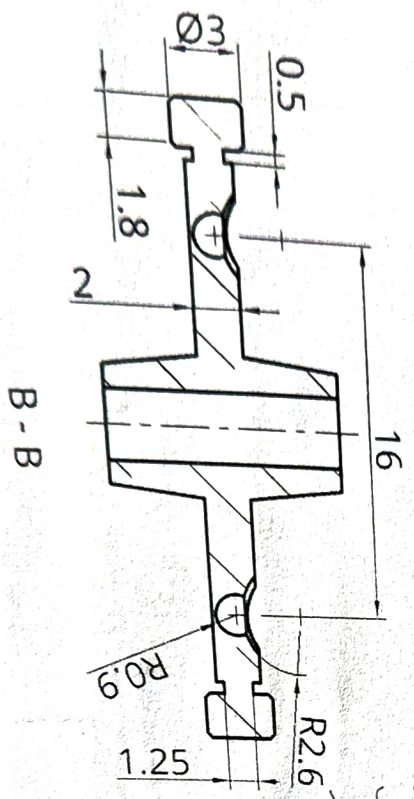


Figure 4
Wobbling Disk
Qty 2

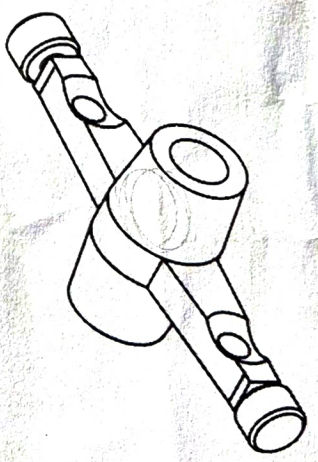


0.2

4.5
4.5
1.0

4.8
4.8
1.6

4.9
4.9
1.8



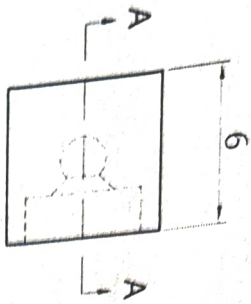


Figure 5
Slider
Qty 4

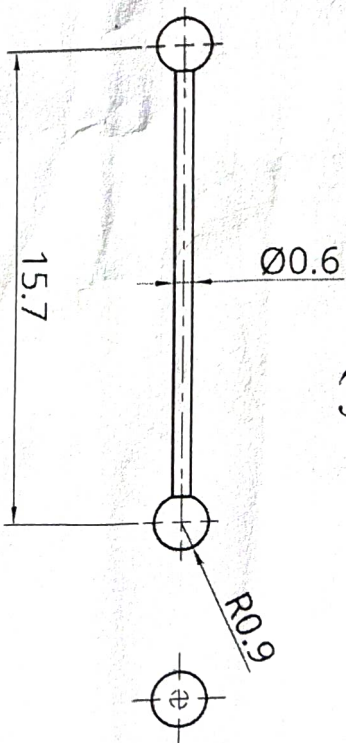
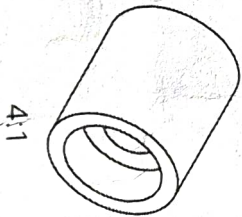
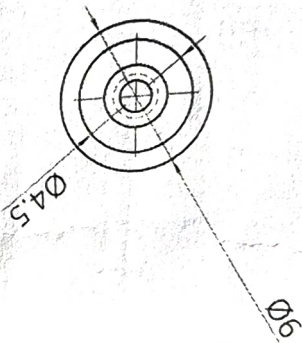
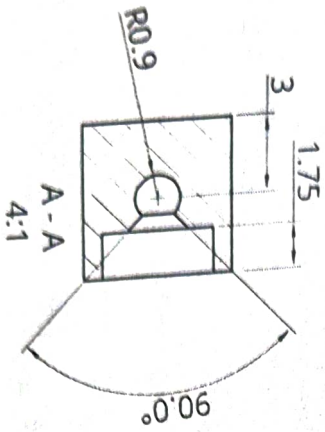
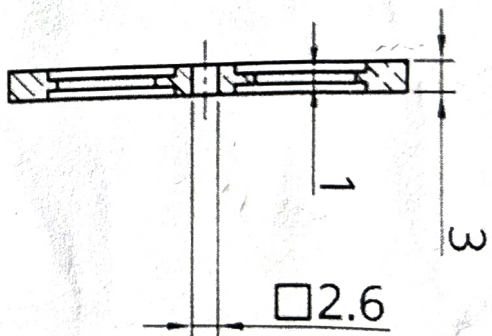
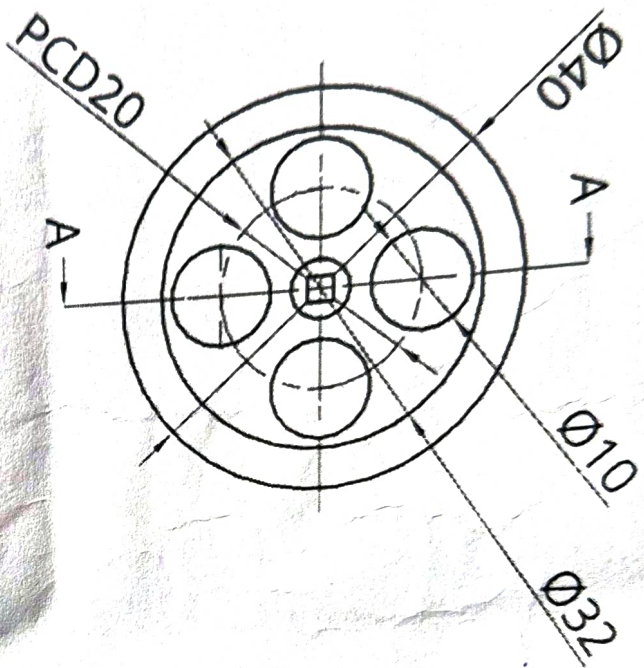


Figure 6
Connecting Rod
Qty 4



A - A

Figure 7

Fly wheel
Qty 2

